

Thank you for purchasing a FANUC Robot.

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- ◆ Straight-head screwdriver ◆ Torque wrench
- ◆ Cross-head screwdriver ◆ Metric hex-head wrenches

Required Items

- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Controller Power Source Label
- ◆ Customer-supplied properly rated input power cable; conductor size of AWG10; terminal size is M5
- ◆ Multimeter ◆ Rigid platform to hold robot
- ◆ 8-M20 x 65mm bolts ◆ Loctite #243
- ◆ 8-M20 flat washers

Optional Items

- ◆ Strain relief cable grip
- ◆ Robot locator pins

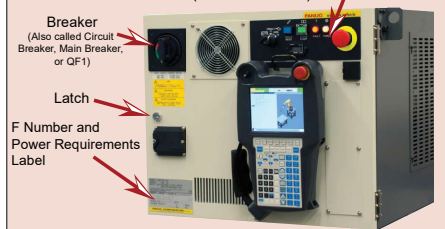
Three Phase

440-500VAC *

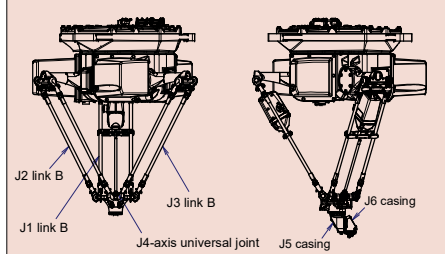


*Dependent on transformer choice.

A-Cabinet Controller (Mate not shown) Operator Panel



Robot (Mechanical Unit) with joints indicated



4-axis (M-2iA/3S, 3SL, M-3iA/6S) 6-axis (M-2iA/3A, 3AL, M-3iA/6A)

A. Unpack and Install your Controller

Shipment Contents



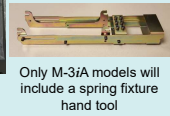
Controller Skid and Robot Skid (might vary)



1. Squeeze and pull the clips off of the skid and remove the cardboard cover. Unwrap the controller and robot. Open all of the boxes included in your shipment.



Teach Pendant, Hardware, & Inspection Data Sheet



Only M-3iA models will include a spring fixture hand tool

Store these items in a safe, known location.

4. Perform Steps 4a through 4j to connect A-Cabinet power. Perform Steps 4k through 4s to connect Mate cabinet power.

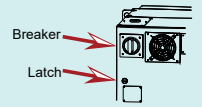
WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

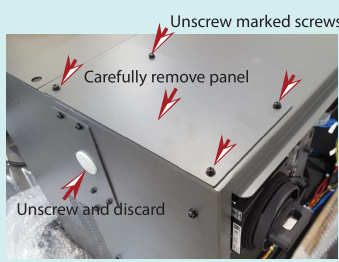
A-Cabinet Controller Power Connection

4a. Determine the power requirements for your controller from the label on the front. The label is at the bottom left corner of the controller door.

4b. Open the controller door by using a straight-head screwdriver to turn the latch clockwise, then turn the Breaker counter-clockwise past its off position.



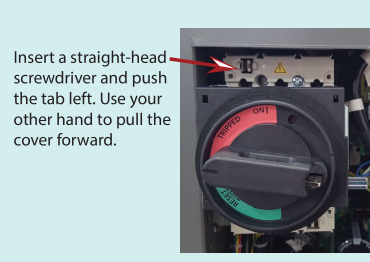
4c. Unscrew the round white plug from the side of the controller. Remove the four screws highlighted in the illustration below, and carefully remove the breaker access panel. It may stick as it has a sealant around the edges.



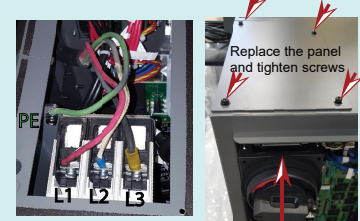
4d. Route the user-supplied input power cable through the side hole. It is typical to purchase and install a strain relief cable grip (example shown below) to remove stress on the power cable connections and isolate the power cable from the controller cabinet.



4e. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.

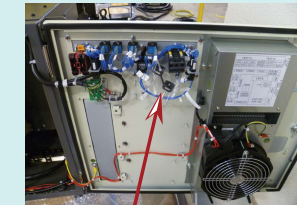


4f. Verify that the input power cable is not energized. Connect L1, L2, L3, and ground to the terminals similar to what is shown on the left below.



Insert cover

4i. Before you close the controller door, cut loose the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.



Key location inside controller

4g. Re-insert the breaker cover and click it into place. Place the breaker access panel back over top of the breaker and secure with four screws.

4h. With the breaker still off, energize the input power cable and use a multimeter to verify the correct voltage is present via the holes in the breaker cover.

4j. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.

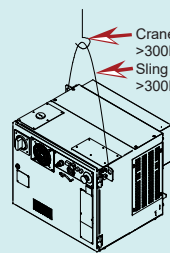
Controller installation is complete. DO NOT TURN ON THE CONTROLLER UNTIL YOU ARE INSTRUCTED TO DO SO. Go to B. Unpack and Install your Robot.

2. Identify the controller and use properly rated equipment to remove the controller from its skid.

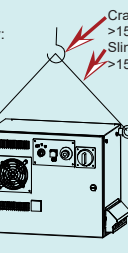
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

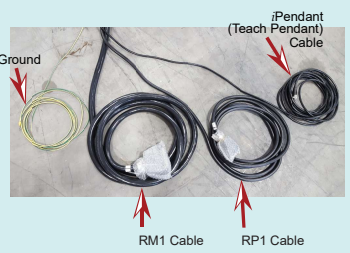
A-Cabinet Controller



Mate Controller

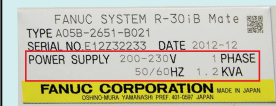


3. Place the controller in a safe location where the grounding wire, RM1, and RP1 cables can reach the robot, and the controller is safely reachable from outside of the robot work area.

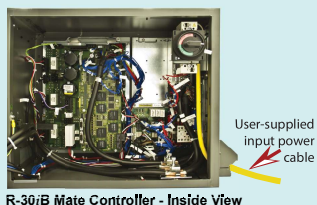


Mate Controller Power Connection

4k. Find the label on the bottom right corner of the controller door. Confirm the power source needed for your controller.

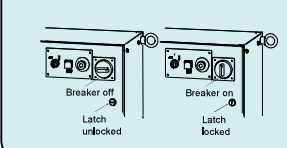


4m. Route the user-supplied input power cable through the side of the controller.

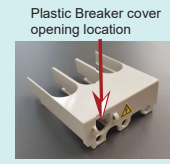


R-30iB Mate Controller - Inside View

4l. Open the controller door by using a straight-head screwdriver to turn the latch counter-clockwise, then turn the Breaker counter-clockwise past its off position.

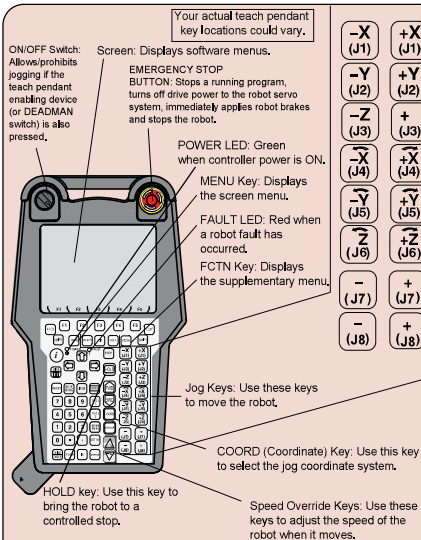


4n. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.



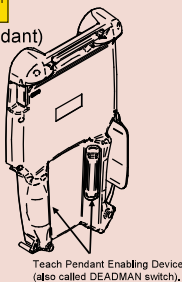
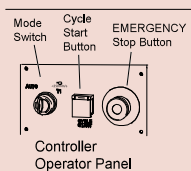
Go to Page 2

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.



Refer to your Application-Specific Setup and Operations Manual for more information on each teach pendant key.

iPendant (Teach Pendant)



For more information

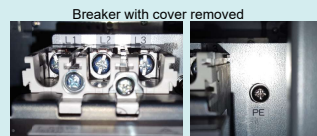
To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/>.



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4o. Connect power to the Breaker with the customer-supplied input power cable. Connect L1 to 1, L2 to 3, and L3 to 5. Connect the ground to the plate located to the right of the Breaker labeled PE.



4p. After the power cable is connected, reinsert the plastic cover.

4q. Use a multimeter to verify the correct voltage is present.

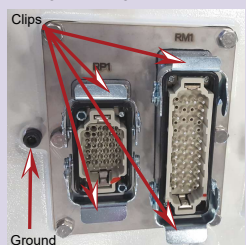
4r. Before you close the controller door, retrieve the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.



4s. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller clockwise.

Controller installation is complete. **DO NOT TURN ON THE CONTROLLER UNTIL YOU ARE INSTRUCTED TO DO SO.** Go to **B. Unpack and Install your Robot.**

7. Connect the green ground wire, and secure with the screw. Connect RP1 and RM1 then secure with clips. These cables are also referred to as the Robot Connection Cable.



8. Connect the teach pendant to its cable as shown below. Rotate the teach pendant cable until it is seated. Then, spin the end of the connector until it is threaded on tight. Wiggling the connector might let you tighten it more.



Robot installation is complete. You are now ready to go to **C. Turn on your Controller.**

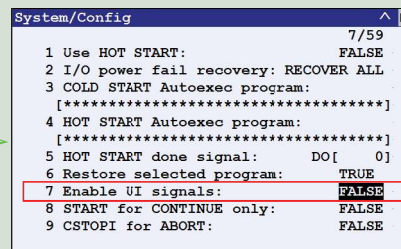
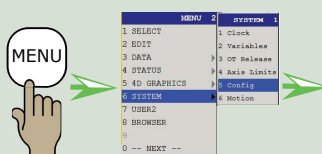
D. Jog your Robot

13. To jog the robot in JOINT mode:

a. Turn the Teach Pendant ON/OFF switch to the ON position. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.

b. Press MENU, select 0 (Next), select (SYSTEM), and then select CONFIG.

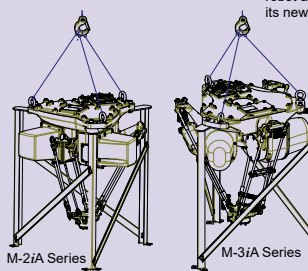
- Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.
- Use the teach pendant F5 key to set the option to FALSE.



B. Unpack and Install your Robot

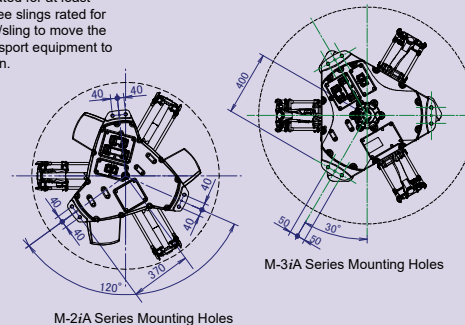
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.



5. The robot is mounted to transport equipment on the skid. Remove the bolts holding the transport equipment to the skid. Use a crane rated for at least 300kg and three slings rated for at least 250kg/sling to move the robot and transport equipment to its new location.

6. Fasten the robot to the fixture with (6) M20 bolts (tightening torque 98NM (10kgfm)). More information is available in the controller specific *Mechanical Unit Operator's Manual*.



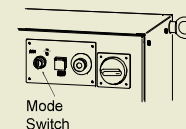
C. Turn on your Controller

WARNING

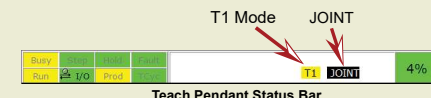
Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

9. Turn on the controller by rotating the Breaker clockwise to ON.

10. Locate the Mode switch on the Controller Operator Panel. Use the keys retrieved in Step 4 to switch to T1 mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.



11. Verify the controller is in T1 Mode by looking at the teach pendant screen. The mode indicator will be highlighted yellow and look similar to the status bar below. If it is not T1, repeat the previous step.



12. Press the COORD key on the teach pendant until JOINT is displayed similar to the status bar above.

You have turned on your controller. You are now ready to go to **D. Jog your Robot.**



c. Use the teach pendant arrow keys to move the cursor to the Hand Broken option (around line 42) and press ENTER.

Hint: You can hold the SHIFT key and press the down arrow key to cursor through the list faster.

- Use the teach pendant F5 key to disable the hand broken error for each motion group.

d. Press and hold a DEADMAN switch on the back of the teach pendant to its middle position. Press RESET on the teach pendant to reset any existing alarms.

NOTE: To view alarms, press MENU, cursor to ALARM, and press ENTER. To toggle between alarm history and active alarms, press F3.

e. To move each joint axis of the robot (called "jogging the robot in JOINT mode"), press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the **Robot (mechanical unit) with joints indicated** diagram on Page 1 of this guide.

Try jogging each joint of the robot.

Turning on and Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the *Controller-specific Software Installation manual*.